TECH STACK

Stochastic: an AI special-situations financing team (proposed stack, not final)

FRONTEND

Next.js · React · TypeScript · Tailwind

Reviewer console: deal board, escalation cards, side-by-side plan table, audit log viewer.

BACKEND

Python · FastAPI · NumPy

OR-Tools (optional)

Workflow engine (state machine and gateway), deterministic deal math, seeded Monte Carlo, quote and number validator.

DATA

Postgres on Tiger Cloud

pgvector / pgvectorscale · Zero-copy forks

Exact facts in tables (deals, clauses, decisions, policies). Vectors only find text. Audit log is append-only and hash-chained.

MODELS

Claude (Anthropic API) · Gemini (optional)

Models read, reason, and write memos. They never produce deal numbers. Code does the math.

PUBLIC DATA AND HOSTING

SEC EDGAR · NY Fed SOFR · FRED · Census · Damodaran · Vultr

Agreements and financials come from EDGAR, cached before the demo (no SEC calls on stage). Rate and volatility inputs are labeled assumptions.

TRACKS

MAIN **Hyphenate** - autonomous finance system, $8K / $5K / $3K plus interviews.

REAL FIT **Tiger Data** (MLH) - Postgres forks hold alternative structures.

CHEAP **Vultr** (MLH) - host the backend and API.

CHEAP **Gemini API** (MLH) - only if a real step uses it.

CHEAP **GoDaddy Registry** (MLH) - register after kickoff.

ASK **Microsoft** - finance named, rules on required products unknown.

THIN **CRS** - consumer credit data, weak fit.

SKIP Bright Data (bans mocked data), Apify, Coinbase, Fetch.ai, Creao, Convex, Spacetime.

Anthropic gives hackers $50 in API credits. We claim no Anthropic prize.

Logos left out on purpose (text chips only). Prize rules are unconfirmed until the Live Site and booths confirm them. MLH prizes need an opt-in on Devpost.

WORKFLOW

One master agent, four specialists, one gateway, and a human who is asked at the moment it matters

1. INPUT

Deal documents (credit agreement, amendments, financials, term sheets) plus a one-line brief: amount, purpose, timeline, hard limits.

EVENTS ARRIVE DURING THE DEAL
rate move - lender changes terms - late document - reviewer feedback

DEAL MANAGER (MASTER AGENT)

Classifies the situation (8 types), picks a playbook, creates tasks, marks stale results when an event lands, decides when to ask the human.

Intake

Sorts and parses documents. Builds a fact table with a citation for each fact. Lists missing documents.

E1 E2

Clause reader

Finds baskets, covenants, and definitions. Builds the constraint set with exact quotes and confidence.

E3

Structuring

Builds mixes of senior, junior, and preferred under the constraints. Keeps the best trade-offs.

E4

Risk

Runs seeded Monte Carlo on the shortlist. Reports breach, dividend, and loss odds.

E5

TASKS AND RESULTS ARE TYPED WORK ITEMS IN POSTGRES. AGENTS DO NOT CHAT WITH EACH OTHER.

MEMO AND APPROVALS AGENT

Writes the recommendation and the side-by-side table. Turns a rejection into a proposed policy with evidence.

E6

GATEWAY (CODE)

The only writer. Checks permissions, blocks self-approval, logs every action and every denied attempt.

HUMAN REVIEWER

A senior person who has done this work. Confirms key facts, answers questions, approves, rejects, or asks for changes.

MEMORY

Past deals, decisions, and approved policies feed the next deal.

POSTGRES ON TIGER CLOUD work_item | fact + citation | document_chunk (pgvector) | precedent | policy | audit_log (hash-chained) | forks for what-if plans

E1 to E6 mark where an agent can stop and ask the human. See the next page. The workflow follows the Hyphenate workshop blueprint (NEEDS_INFO, APPROVED, POSTED states).

ASK EARLY. PROVE IT.

The human is asked when the agent is unsure, not only at the end

WHEN AN AGENT ASKS

- E1 **Missing document.** The existing agreement or an amendment is not in the upload. The agent requests it and pauses that part.
- E2 **Facts to rely on.** One screen of the facts that drive the plan. The reviewer confirms or corrects them.
- E3 **Low-confidence clause.** The quote is unclear or a defined term cannot be resolved. The agent asks and does not guess.
- E4 **Plan is blocked.** A clause stops the first plan. The agent shows the redesign, the consent or waiver route, and the price of the clause.
- E5 **Recommendation flips.** A rate move or lender change changes the best plan. The reviewer sees what changed and why.
- E6 **Final approval.** The reviewer approves, rejects, or asks for changes. A rejection becomes a proposed policy.

PROOF FOR JUDGES

- **Real deal, answer key.** Replay a real public deal (Organogenesis, Nov 2024) from the documents that existed before it. Compare with what actually happened.
- **Catch-rate score.** Planted and real clauses with recall, precision, and silent-error counts.
- **Deterministic math.** 14 closed-form tests for the simulator. No number reaches a memo without a run id.
- **Limits in code.** The proposing agent cannot approve its own plan. Only the gateway posts.
- **Adversarial tests.** Self-approval, a forged amendment, a prompt injected inside the loan agreement, a swapped plan after approval, a poisoned policy.
- **Honest framing.** A reconstruction from public filings, not the lenders' internal views.

REAL ENOUGH?

Real filings, real clauses, and an answer key from what actually happened.

WOULD YOU HIRE IT?

It asks early, limits are enforced in code, and every step is in the audit log.

SOMETHING NEW?

Planning the consent or waiver route when a clause blocks a plan. Our research found no product for it, which is not proof that none exists.

THE MOMENT

The agent catches the clause that blocks the plan, re-plans, and defends the new plan with evidence.

Design proposal. No practitioner has checked the assumptions yet, including whether redeemable preferred counts as debt under a given agreement.